

Kapture Finance — Maya Collections Voicebot

High-Level Design (HLD) • AI Delivery Intern Take-Home Assignment

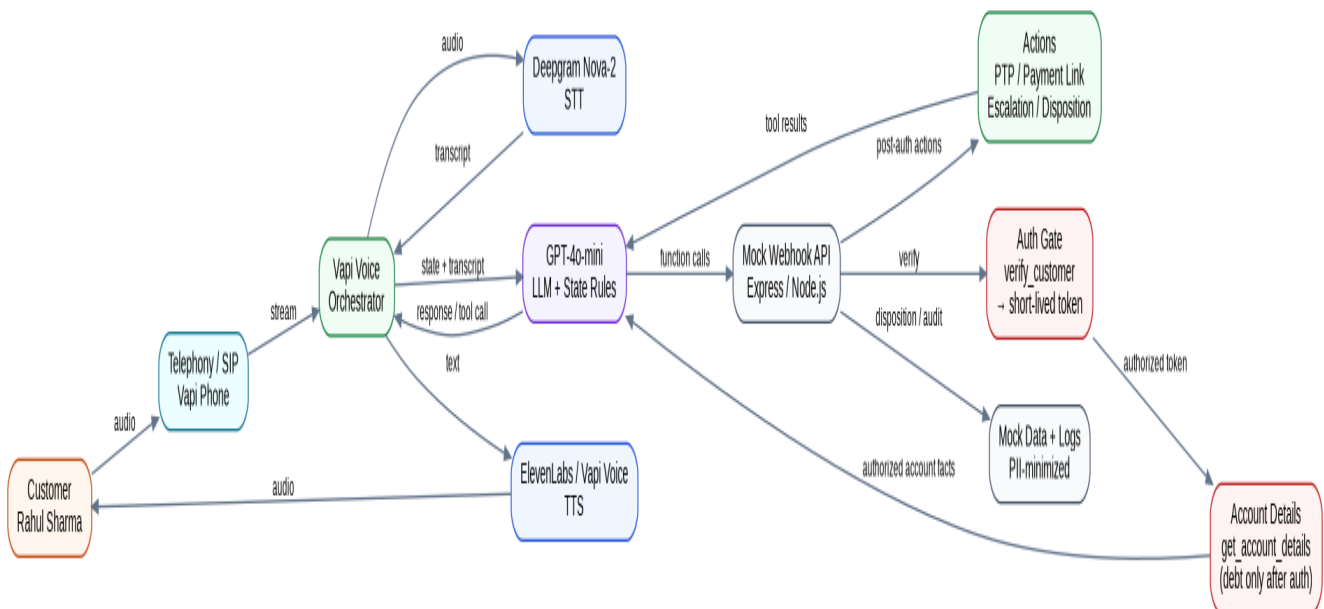
Agent: Maya **Client:** Kapture Finance (fictional) **Scenario:** outbound overdue-EMI collections

Demo context: Rahul Sharma • Personal loan • ₹8,499 overdue • 12 days past due.

Design principle: authentication is a server-enforced security boundary, not merely an LLM instruction. Debt values are retrieved only after successful verification.

1. Architecture & Pipeline

Customer audio enters through telephony, is transcribed by Deepgram, passed to the Vapi orchestration layer and GPT-4o-mini, then synthesized back to speech. Tool calls go to a mock Express webhook that enforces authorization and records dispositions.



Latency budget

Hop	Target	Notes
STT	~200 ms	Streaming transcript / endpointing
LLM first byte	~400 ms	GPT-4o-mini, low temperature
TTS	~300 ms	Streaming synthesis
Network / orchestration	~200 ms	Vapi + tool/network overhead
Total	~1.1 s	Below the assignment's <1.2 s target

The latency figures are engineering targets, not guaranteed SLAs. Tool calls can add latency, so the agent should keep responses concise and avoid unnecessary sequential calls.

2. Conversation Flow / State Machine

State	Purpose	Transition lock
INIT	Call connected	Moves to AUTH_PENDING
AUTH_PENDING	Identity confirmation + verification	Only verify_customer: verified=true
AUTHENTICATED	Authorized account retrieval	get_account_details must return authorized=true
NEGOTIATION	Intent identification	Explicit customer intent
ACTION_EXECUTION	PTP/payment/escalation	Tool result
ESCALATED	Human handling	Transfer/end
CALL_ENDED	Terminal state	Disposition logged

Security invariant: the assistant cannot unlock the disclosure state from conversation alone. verify_customer issues a short-lived authorization token; get_account_details refuses requests without a valid token.

```

AUTH_PENDING
  |
  | verify_customer
  v
verified=false ----> retry / end
  |
  | verified=true + auth_token
  v
AUTHENTICATED
  |
  | get_account_details(auth_token)
  v
NEGOTIATION

```

3. Intents & Entities

Intent	Example	Entities	Route
Confirm_Identity	Yes, Rahul speaking.	verification	Auth
Promise_To_Pay	I'll pay Friday.	PTP_Date, PTP_Amount	PTP
Already_Paid	I paid yesterday by UPI.	payment date/mode/ref	Disposition
Hardship_Claim	I cannot pay the full amount.	Hardship_Reason	Escalate
Dispute_Debt	I dispute this amount.	notes	Escalate
Request_DNC	Stop calling me.	—	Immediate DNC
Wrong_Person	Rahul is not here.	availability	End / log
Callback_Request	Call tomorrow evening.	date/time	Log callback
Hostile	Stop bothering me!	—	Warning → end

4. Tools / API Contracts

Tool	Purpose	Security rule
verify_customer	Verify identity	Allowed in AUTH_PENDING
get_account_details	Return account/debt facts	Requires auth_token
log_promise_to_pay	Record explicit PTP	Requires auth_token
send_payment_link	Mock SMS/WhatsApp payment link	Requires auth_token
escalate_to_agent	Human route for dispute/hardship	Requires auth_token
mark_disposition	Final call outcome	Pre-auth DNC/wrong-person allowed; post-auth statuses require token

The complete JSON schemas are supplied separately in vapi/tool_definitions.json.

5. Authentication & Data Safety

- Before authentication, the assistant may identify itself and ask whether it is speaking with Rahul, but it must not disclose or confirm debt existence, loan/EMI details, amount, overdue status, balance or DPD.
- A successful verify_customer call creates a short-lived authorization token. get_account_details is the only path to the demo debt facts.
- Production tokens should be signed, short-lived and bound to the call/session. Verification secrets and tokens must never be logged.
- PII should be masked in logs and sensitive data encrypted in transit and at rest.

6. Guardrails & Compliance

- Use the assignment's 08:00–19:00 local calling window.
- Immediate DNC handling; no further negotiation after a DNC request.
- No threats, harassment, intimidation or shaming.
- No debt disclosure to third parties or voicemail.
- No invented fees, waivers, settlements, extensions or legal/credit consequences.
- Tool failures must not be replaced with guessed information.
- Keep the agent scoped to the collections workflow and preserve the same security rules during English/Hindi switching.

7. Edge Cases

Case	Expected behavior
Already paid	Ask payment date/mode; log ALREADY_PAID; do not demand another payment.
Dispute	Acknowledge; escalate to human resolution; log DISPUTED.
Hardship	Empathy; escalate; do not invent concessions.
Wrong person	No debt disclosure; log WRONG_PERSON if appropriate.
DNC	Log DO_NOT_CALL immediately and terminate.
Voicemail / silence	No debt disclosure; limited re-prompts; NO_RESPONSE.
Hostile	One calm warning; terminate if abuse continues.
Language switch	Switch EN↔HI while preserving state and auth rules.
Tool failure	Do not hallucinate; apologize and escalate/end safely.

8. Escalation, Disposition & Observability

Escalate: disputes, hardship requiring non-standard arrangements, explicit human requests, or repeated service failures.

Dispositions: PTP_AGREED, ALREADY_PAID, DISPUTED, HARDSHIP_ESCALATED, WRONG_PERSON, DO_NOT_CALL, NO_RESPONSE, HOSTILE, CALLBACK_REQUEST, AUTH_FAILED.

Metric	Definition
Containment Rate	Calls resolved without human escalation
PTP Rate	Calls ending in a valid Promise-to-Pay
FCR	Calls with a valid final disposition
Avg Response Latency	Average voice response latency
Tool Failure Rate	Failed tool calls / total tool calls
Drop Rate	Unexpectedly disconnected calls

Metric	Definition
Auth Success Rate	Successful verification / verification attempts
Debt Leakage Rate	Unauthorized debt disclosures; target = 0

Trace each call as: call_id → state → transcript turn → tool call → tool result → response → disposition. The most important compliance alarm is Debt Leakage Rate > 0.

9. Demo Acceptance Criteria

Happy path: Greeting → identity confirmation → verification → account retrieval → debt disclosure → PTP → payment link → disposition → close.

Edge path: Greeting → verification → account retrieval → already-paid/dispute/DNC → correct disposition → close.

10. Future Improvements

- Replace the mock server with authenticated lending APIs and a real DNC service.
- Bind authorization to Vapi call IDs and add signed webhook authentication.
- Add real SMS/WhatsApp and human transfer integrations.
- Build automated prompt-injection and debt-leakage red-team tests.
- Add English/Hindi evaluation sets and production dashboards.